Blue-print of CDP model from Brands across PAR

CDP as a product can be though of to have four main components:

1. Data Ingestion capability
2. Data export capability
3. Data model that calculates and processes the mapping
4. UI to add, modify and remove specific data sources and desired user attributes

This page expands on point 3 - Data model that calculates and processes the mapping, specific to data sources from PAR brands. It provides information on the CDP attributes that can be gleaned from the various data sources across PAR and how they can connected to build meta model of profiles.

We considered three main data sources in arriving at this CDP model:

- PAR Loyalty
- PAR Pay
  - Infinicept
  - Aurus
- PAR POS
- PAR Ordering

The CDP attributes are divided into five types:

- Identity Meta Attributes
- Transaction Behavior Attributes
- Loyalty Behavior Attributes
- Engagement Behavior Attributes
- Predicted Meta Attributes

The most granular data is at a PROFILE level. A profile could either a specific email, a specific user_id, a specific card profile, or a specific combination of meta attributes put together (to keep it generic).

PAR Loyalty

- PAR Loyalty currently has **Email Address** for all Loyalty and eClub members
- Furthermore, PAR Loyalty assigns a unique integer **user_id** to each unique Business | Email Address combination (the same Email Address can appear in multiple Businesses and we keep them separate)
- **user_id** is captured in all Loyalty user transactions

(All the attributes highlighted in green are readily available. The ones in red are to be calculated with further transformations)

Pillar	Attributes	User Identifier for Joining to Profile
Identity Meta Attributes	• First Name • Last Name • Age • DOB • Anniversary • Email • Secondary Email • Phone • Gender • Signup Date • Email Subscription Status • PN Subscription Status • SMS Subscription Status • Device • Last Signin at • Last Device used • Signup Channel • Uses Facebook • Preferred Language • Active Referrer • Favorite Location • Most frequented Location • Time zone • Address	Nearly all of these attributes come from PAR Loyalty's **Users** table and form the basis of a User Profile. All rows in the Users table have • **User ID** • **Email Address or Phone**

	• City • State • Country • Additional Profile fields • Migrated from another platform	
Transactional Behavior Attributes	• Total loyalty transactions • Total loyalty amount spent • Offline transaction % • Online transaction % • Avg. check size (loyalty) • Median check size (loyalty) • Days between transactions (loyalty) • Total Rewards received • Total redemptions successfully made • Available Rewards • Channel of Redemption • Favorite type of offer • Favorite time of the day for transaction • Favorite day of the week for transaction • Eats alone vs. Group eater • Favorite breakfast item • Favorite meal item - main course • Favorite meal item - modifiers • Days since last loyalty transaction • Days since last redemption • RFM	All Loyalty Transactions have **Business ID | User ID**

Loyalty Behavior Attributes	• Total loyalty points received • Total non-loyalty points received • Points converted to rewards • Loyalty rewards redeemed • Days between earning loyalty reward and redeeming it • Membership levels • Days since last loyalty points earned • Current available points	All data points present in PAR Loyalty's production tables
Engagement Attributes	**Campaigns** • Campaigns user been part of • Campaigns user participated in • Days since last campaign message was delivered • Days since last campaign awarded points • Days since last reward/coupon delivered **Emails** • Emails sent • Emails opened • Emails clicked • Email Open Rate • Email CTR • Favourite time of day to interact with emails • Email interaction device **Push Notifications** • PNs sent • PNs opened • PN open rate • PN Open mode (read through app, opened from notification)	All Engagement Attributes come with **Business ID | User ID** Furthermore, Email engagement comes with Email Address

	• Favorite time of day to interact with PNs **Referrals** • Referrals sent • Number of Referrals accepted **Feedback** • Feedbacks given • Positive ratings • Negative ratings • Feedback Sentiment Customer Service Food Quality Ambience Wait Time App Experience **Others** • Surveys participated • Active Gift card user • Amount transacted using Gift card • Active Apple Payment User • Active Subscriptions User • Fraudulent Activities	
Predicted Attributes	**Predicted Spend** • Next 30 days • Next 60 days • Next 90 days • Predicted CLV	All Calculated Attributes come with **Business ID | User ID**

PAR Pay

Infinicept

- Infinicept has transaction level records which has card holder identifiable data

- Each card holder profile itself can be treated as entry into the CDP
- In addition, a card holder profile can be tied back to a specific user_id on PAR Loyalty using **PAR POS's transaction ID as the underlying link**

Pillar	Attributes	User Identifier for Joining to Profile
Identity Meta Attributes	• Card first 6 • Card last 4 • Card expiry date • Issuing Bank • Card Network (Visa/Mastercard) • Card Type (Credit/Debit)	Card first 6 Card last 4 Card expiry date
Transaction behavior Attributes	• Total card based transactions • Total amount spent on card based transactions • Credit Card % • Debit Card % • Favorite time of the day for card based transactions • Favorite day of the week for card based transactions	Card first 6 Card last 4 Card expiry date

PAR Pay

Aurus

- Aurus has transaction level records which has card holder identifiable data i.e, Card holder name
- It has one to one mapping with the transactions from Infinicept

Pillar	Attributes	User Identifier for Joining to Profile

Identity Meta Attributes	• Card Holder Name	Transaction ID mapped to Transaction ID of Infinicept, which is then mapped to card holder profile or user_id from PAR Loyalty

PAR POS

- PAR POS has transaction level data
- It has one to one mapping with transactions from PAR Pay (Infinicept) and PAR Loyalty

Pillar	Attributes	User Identifier for Joining to Profile
Transaction Behavior Attributes	• Channel of Transaction (Dine In, Doordash etc.) • Tip Amount	PAR POS's Order ID mapped to PAR Loyalty's Order ID and Transaction ID of PAR Pay - Infinicept

PAR Ordering

- PAR Ordering has transaction level data, for all the orders placed on the PAR Ordering application
- Each order has the ability to capture user information (listed below) along with geographical information, which is not found in the data sets of any other PAR brand data sources
- Some users have loyalty accounts, for which points related information is stored
- The exact attribute(s) based on which the order can tied back to PAR Loyalty, PAR POS still needs to be explored

Pillar	Attributes	User Identifier for Joining to Profile
Identity Meta Attributes	• First Name • Last Name • Email • Phone • DOB	The source of user meta information is mainly from Customers, Customer_Demographics

	• Identification Number (Govt.)
	• User Type (Customer, Guest)
	• Profile Creation Date
	• Email Subscription Status
	• PN Subscription Status
	• SMS Subscription Status
	• Device Type of Profile Creation (Signup Channel)
	• Device UUID
	• Preferred Language
	• Active Referrer
	• No.of Referral made
	• Most frequented Location
	• Time zone
	• Address (with door no., and can be more than one)
	• City
	• State
	• Country
	• Latitude
	• Longitude
	• Post Code
	• Vehicle Information
	  ◦ Brand
	  ◦ Model
	  ◦ License Plate Number (last 3 digits)
		tables and form the basis of a User Profile.
		
		All rows in these User related tables have
		
		• **Customer ID**
		• **Email Address or Phone**

Transactional Behavior Attributes	• Total transactions • Total amount spent • Avg. check size (Total) • Loyalty transactions • Total loyalty amount spent • Avg. check size (Loyalty) • Transaction count by Order Type (Dine-In, Take Out, Curb-side,	

Delivery)

- Total amount spent by Order Type (Dine-In, Take Out, Curb-side, Delivery)
- Avg. check size by Order Type (Dine-In, Take Out, Curb-side, Delivery)
- % Transaction by Order Type (Dine-In, Take Out, Curb-side, Delivery)
- % Total amount spent by Order Type (Dine-In, Take Out, Curb-side, Delivery)
- Total Discounts
- Avg. Discount amount per transaction
- Total Discounts by Order Type
- Avg. Discount amount by Order Type
- % of Transactions with Discounts
- % of Transactions by Discount Channel (POS, Mobile, Kiosk etc.)
- % of Transactions by Discount Type (Coupon, Loyalty Reward)
- Favorite offer type
- Avg. No.of items ordered per Order
- Avg. No.of items ordered per Order by Item Category (main course, side dish etc.)
- % of Transactions by Day Part (Hours)
- % of Transactions by Week Part
- Eats alone vs. Group eater
- Favorite breakfast item
- Favorite lunch item
- Favorite dinner item
- Favorite meal item - main course
- Favorite meal item - modifiers
- % of Transactions by Payment Method (Online, COD)

	• % of Transactions by Payment Processor (Toast, Aurus etc.) • % of Transactions by Payment Card Type (VISA, Mastercard etc.) • % of Transactions that has tax included • % of Transactions that has tip • Total Tip amount • Avg. Tip amount per transaction • Most Frequent tax % levied • Avg. Distance of ordering location from the delivery address • % of Transactions that are cancelled • % of Transactions that were returned • Avg. Time Diff in secs between order completion and pickup • % Transactions by Partner delivery channel (Doordash, Grubhub, Uber Eats etc.) • % of Transactions by Delivery schedule type (Scheduled, ASAP)	
Loyalty Behavior Attributes	• Total points received • Total available points	

(By default I applied black color instead of green or red, because I haven't gotten around to write a sql to get the actual data)

Example:

Here is a real example gleaned from a user profile belonging to Salsarita:

Data Source	Pillar	Column Name	Example Column Value

PAR Loyalty	Identity Meta Attributes	Email	tr*********t@aol.com (masked to protect PII)
		First Name	T***y (masked to protect PII)
		Last Name	R*****t (masked to protect PII)
		Birthday	2003-Aug-20
		Anniversary	NA
		Age (in years)	19
		Secondary Email	NA
		Phone	50*******5 (masked to protect PII)
		Gender	NA
		Signup Date	2019-Mar-01
		Period on Platform (days)	1482
		Email Subscription Status	Yes
		PN Subscription Status	Yes
		SMS Subscription Status	No
		Opted for e-Receipt	No

		Banned or Deactivated	No
		First Device	iPhone 8
		OS	iOS 12
		Last Signin At	2021-July-11
		Last Device Used	iPhone 12
		Signup Channel	MobileEmail
		Uses Facebook	No
		Preferred Language	EN
		Active Referrer	No
		Favorite Location	Middletown - 135
		Most Frequented Location	Middletown - 135
		Time zone	America/New_York
		Address	12915 Shelbyville Rd.
		City	Louisville
		State	KY
		Country	USA
		Additional Profile Fields	NA
		Migrated from another Loyalty Platform	No
PAR Loyalty	Transaction Behavior Attributes	Total Loyalty Transactions	25
		Total Loyalty Amount Spent	$606.2

		Offline Transaction %	92%
		Online Transaction %	8%
		Avg. Loyalty Check Size	$24.25
		Median Loyalty Check Size	$23.96
		Avg. Days Between Loyalty Transactions	59
		Total Rewards Received	171
		Total Successful Redemptions	14
		Available Rewards	1
		Channels of Redemption	Online - 7% Mobile or POS - 93%
		Favourite Type of Offer	$ Off
		Favorite time of the day for transaction	11AM - 12 PM
		Favorite day of the week for transaction	Sunday
		Consumption Behavior	Single Person Consumption
		Favorite breakfast item	Fountain Drink
		Favorite meal item - main course	Chicken Taco
		Favorite meal item - modifiers	TOPPINGS: Sour Cream

		Days since last loyalty transaction	19
		Days since last redemption	20
		R - Quartile	92%
		F - Quartile	78%
		M - Quartile	72%
PAR Loyalty	Loyalty Behavior Attributes	Total loyalty points received	607
		Total non-loyalty points received	111
		Points converted to rewards	70
		Loyalty rewards redeemed	7
		Days between earning loyalty reward and redeeming it	15
		Membership levels	NA
		Days since last loyalty points earned	19
		Current available points	18
PAR Loyalty	Engagement Attributes - Campaigns	Campaigns user been part of	472
		Campaigns user participated in	28
		Days since last campaign message	2

		was delivered	
		Days since last campaign awarded points	15
		Days since last reward/coupon delivered	7
PAR Loyalty	Engagement Attributes - Email	Emails sent	
		Emails opened	
		Emails clicked	
		Email Open Rate	
		Email CTR	
		Favourite time of day to interact with emails	
		Email interaction device	iPhone 12
PAR Loyalty	Engagement Attributes - Push Notifications	PNs sent	231
		PNs opened	20
		PN open rate	8.65%
		PN Open mode (read through app, opened from notification)	read_through_app
		Favorite time of day to interact with PNs	11AM - 3PM
PAR Loyalty	Engagement Attributes - Referrals	Referrals sent	0

		Number of Referrals accepted	0
PAR Loyalty	Engagement Attributes - Feedback	Feedbacks given	0
		Positive ratings	0
		Negative ratings	0
		Feedback Sentiment - Customer Service - Food Quality - Ambience - Wait Time - App Experience	NA
PAR Loyalty	Engagement Attributes - Others	Surveys participated	0
		Active Gift card user	No
		Amount transacted using Gift card	0
		Active Apple Payment User	0
		Active Subscriptions User	0
		Fraudulent Activities	0
PAR Loyalty	Predicted Attributes	Predicted Spend - Next 90 days	$51.67
PAR Pay - Infinicept	Identity Meta Attributes	Card first 6	443041
		Card last 4	2748
		Card expiry date	08/2023

		Issuing Bank	PNC Bank National Association
		Card Network (Visa/Mastercard)	Visa
		Card Type (Credit/Debit)	Consumer Debit
PAR Pay - Infinicept	Transaction Behavior Attributes	Total card based transactions	8
		Total amount spent on card based transactions	$206.67
		Credit Card %	0%
		Debit Card %	100%
		Favorite time of the day for card based transactions	11AM - 12 PM
		Favorite day of the week for card based transactions	Sunday
PAR Pay - Aurus	Identity Meta Attributes	Card Holder Name	R*****t/T****j